MIL-E-1/1075C
27 December 1968
SUPERSEDING
MIL-E-1/1075B
21 September 1964

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 5768

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, planar

See figure 1

Mounting position: Any

Weight: 3 ounces nominal

ABSOLUTE RATINGS: CW amplifier

Parameter:	Fi	Ef	Eb	Ib	Pp	Ec	Rk	tk	T (seal)
Unit:	MHz	V	Vdc	mAdc	W	Vdc	Ohms	sec	°C
Maximum:	500	6.9	250	15	2.0	---	---	---	175
Minimum:	---	5.7	---	---	---	---	---	---	---
TEST CONDITIONS:	---	6.3	150	---	---	0	100	30	---

GENERAL:

Qualification - Required

'olding period (MIL-STD-1311) t = 96 hours

Source: https://assist.dla.mil -- Downloaded: 202640-15T14:07Z
Check the source to verify that this is the currenversion before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.65	II	If	375	425	mA
1256	Electrode current (1) (anode)		0.65	II	Ib	3.5	12.5	mAdc
1256	Electrode current (2) (anode)	Ec = -6.5 Vdc	0.65	II	Ib	---	50	µAdc
1306	Transconductance (1)		0.65	II	Sm	7,000	12,000	µmhos
1306	Transconductance (2)	Ef = 5.7 V	0.65	II	Sm	6,000	---	µmhos
Ⓒ 4260	Noise figure	F = 500 MHz	0.65	II	NF	---	12.5	db
Ⓒ 4253	Gain	F = 500 MHz	0.65	II	G	12.0	---	db
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	Eg = 100 Vdc Eb = 500 Vdc	4.0	S3	R	100	---	Meg
1031	Low-frequency vibration	Rp = 10,000 ohms	6.5	S3	Ep	---	75	mVac
1266	Total grid current	Eb = 300 Vdc; Rk = 400 ohms	6.5	S3	Ic	0	-1.0	µAdc
Ⓒ 1231	Emission	Rk = 0; (Eb = Ec)/Ik = 30 mAdc	6.5	S3	Es	---	10.0	Vdc
1316	Amplification factor		6.5	S3	Mu	60	125	---
1331	Direct-interelectrode capacitance	See note 1	6.5	S3	Cgp Cgk Cpk	0.85 1.10 ---	1.20 2.00 0.015	pF pF pF
---	Torque	Applied between anode and cathode without shock (see note 2)	6.5	S3	---	10	---	Lb-in.
---	Seal	T = 175° C (min); t = 16 hours (min) (see note 3)	6.5	S3	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group C	---	---	t	500	---	hrs
---	Life-test end points	Gain Noise figure	--- ---	--- ---	G NF	10 ---	--- 15	db db

NOTES:

1. Test in capacity adaptor in accordance with Drawing 260-JAN.
2. After subjection to the torque specified, the tube shall be tested for total grid current and the value found shall not be in excess of -1.2 µAdc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:07Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

3. This test shall consist of subjecting the tube to the temperature and minimum time specified, after which the tube shall be tested for heater current and the value found shall be within 375 to 425 mA.

The margins of this tube specification sheet are marked with a revision letter (C) to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

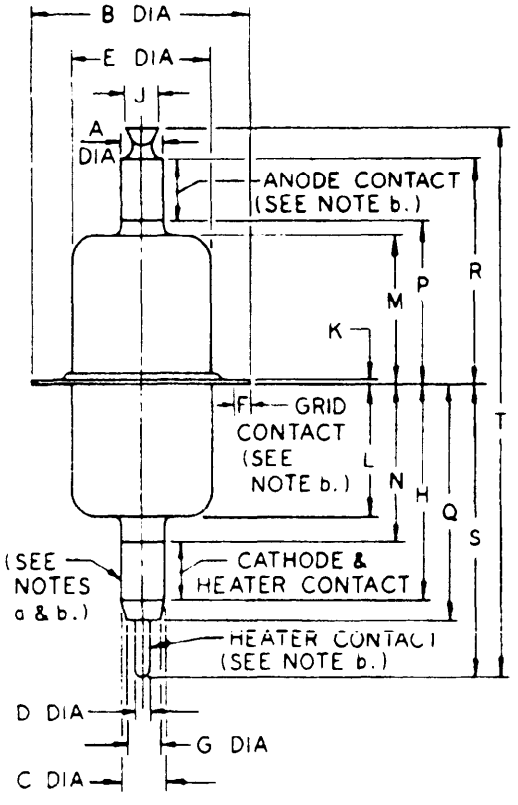
Review activities:
Army - EL
Navy - EC, SH
Air Force - 11, 85
DSA - ES

User activities:
Army - MU
Navy - AS, OS, CG
Air Force - 19

Preparing activity:
Army - EL

(Project 5960-2328-1)

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:07Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 3 (periodic check, see note c)		
A	.197 (5.01)	.207 (5.26)
B	.995 (25.27)	1.005 (25.53)
C	.197 (5.01)	.207 (5.26)
D	.045 (1.14)	.055 (1.40)
E	-- --	.660 (16.76)
F	.080 (2.03)	-- --
G	-- --	.190 (4.83)
H	.825 (20.96)	-- --
J	-- --	.195 (4.95)
K	.007 (.18)	.013 (.33)
L	-- --	.560 (14.22)
M	-- --	.585 (14.86)
N	-- --	.645 (16.38)
P	-- --	.635 (16.13)
Q	.930 (23.62)	.980 (24.89)
R	.865 (21.97)	.915 (23.24)
S	1.145 (29.08)	1.225 (31.12)
T	-- --	2.375 (60.33)

NOTES:
a. Reference diameter 'C' shall be concentric with respective diameters as follows: Dia 'A' .050 (1.27 mm) FIR, Dia 'D' .030 (.76 mm) FIR, Dia 'B' .030 (.76 mm) FIR, Dia 'E' .050 (1.27 mm) FIR. Full indicator reading includes out of roundness and plane tilt of components.
b. Contact areas shall be clean and free from glass.
c. These dimensions shall be inspected periodically every 3 months during production. At start of production, a sample of six tubes selected at random shall meet the quality conformance inspection, part 3, periodic check requirements with no allowable defects. Failure of any dimension to meet the quality conformance inspection, part 3, periodic check requirements, shall cause that dimension to be inspected with an AQL of 2.5 percent individual and 4.0 percent combined. When three successive samples as specified above meet the quality conformance inspection requirements for this dimension, the dimension shall revert to periodic check inspection.

FIGURE 1. Outline drawing of electron tube type 5768.

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:07Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL-TD-SS
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:07Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/1075C Electron Tube, Negative Grid(Microwave) Type 5768

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.
ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:07Z
Check the source to verify that this is the current version before use.